🎾 Global Academy & Smart Court Management System

An enterprise-grade, multi-tenant SaaS Sports Director Operating System designed to manage elite global sports infrastructures, localized facility telemetry, predictive scheduling optimization, and edge-computed analytics. Built on a modern full-stack framework using **.NET Core**, **Entity Framework Core**, and **PostgreSQL**, this system abstracts hyper-localized physical operations while leveraging autonomous **Agentic AI** workflows, local computer vision engines, and high-fidelity speech orchestration matrices to maximize venue utilization, financial integrity, and player progression.

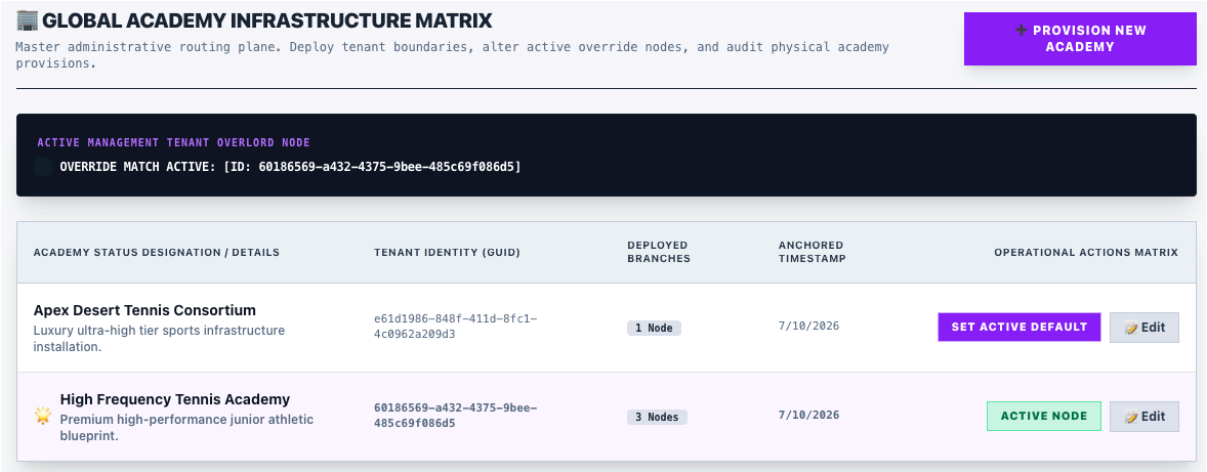
🛠️ Technology Stack & Dependencies

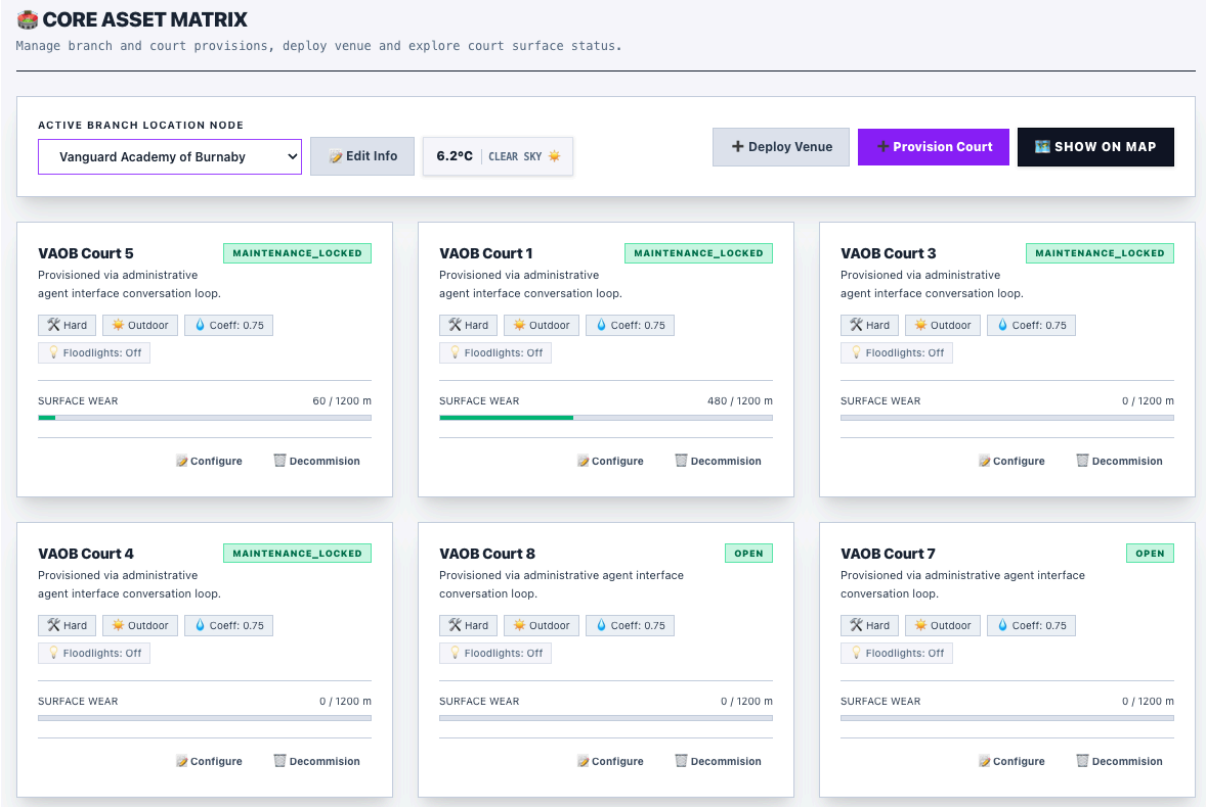
This system relies on free, open-source enterprise frameworks and edge-computed AI libraries:

- **Backend Core Engine:** .NET Core (v8/v9/v10) with ASP.NET Core Web APIs.
- **Database Management:** PostgreSQL Object-Relational Database Server.
- **Object-Relational Mapper:** Entity Framework Core with Npgsql.
- **Security & Authentication:** Microsoft.AspNetCore.Identity.EntityFrameworkCore and JwtBearerextensions.
- **AI Orchestration Framework:** Microsoft.SemanticKernel Core Package, LangGraph, RAG.
- **Third-Party Payment SDKs:** Official Braintree and Square .NET SDK NuGet assemblies.
- **Python Analytics Microservice:** FastAPI + Uvicorn asynchronous task runner engine.
- **Computer Vision & Physics:** Ultralytics YOLOv11, OpenCV (cv2), and NumPy.

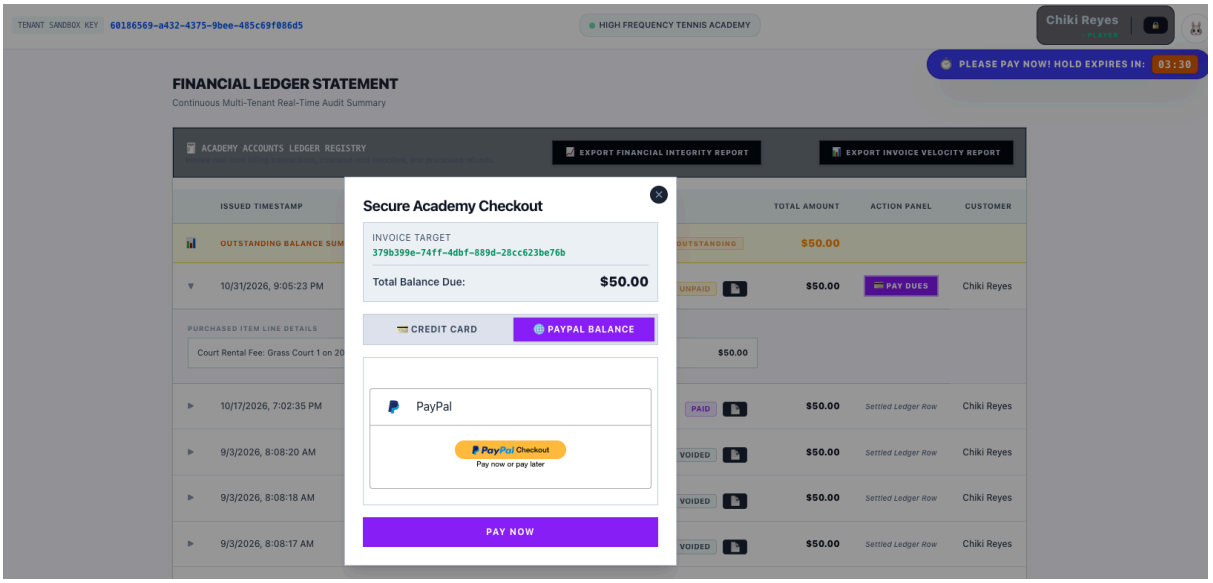
- **Local Speech & Voice Tools:** Faster-Whisper (CTranslate2) and asynchronous Edge-TTS.
- **Data Visualization & PDF Core:** Matplotlib (Agg), Seaborn, and ReportLab.
- **Frontend App Architecture:** Vite + React SPA with Axios (apiClient), Tailwind CSS, and HTML5 Media/Audio APIs.

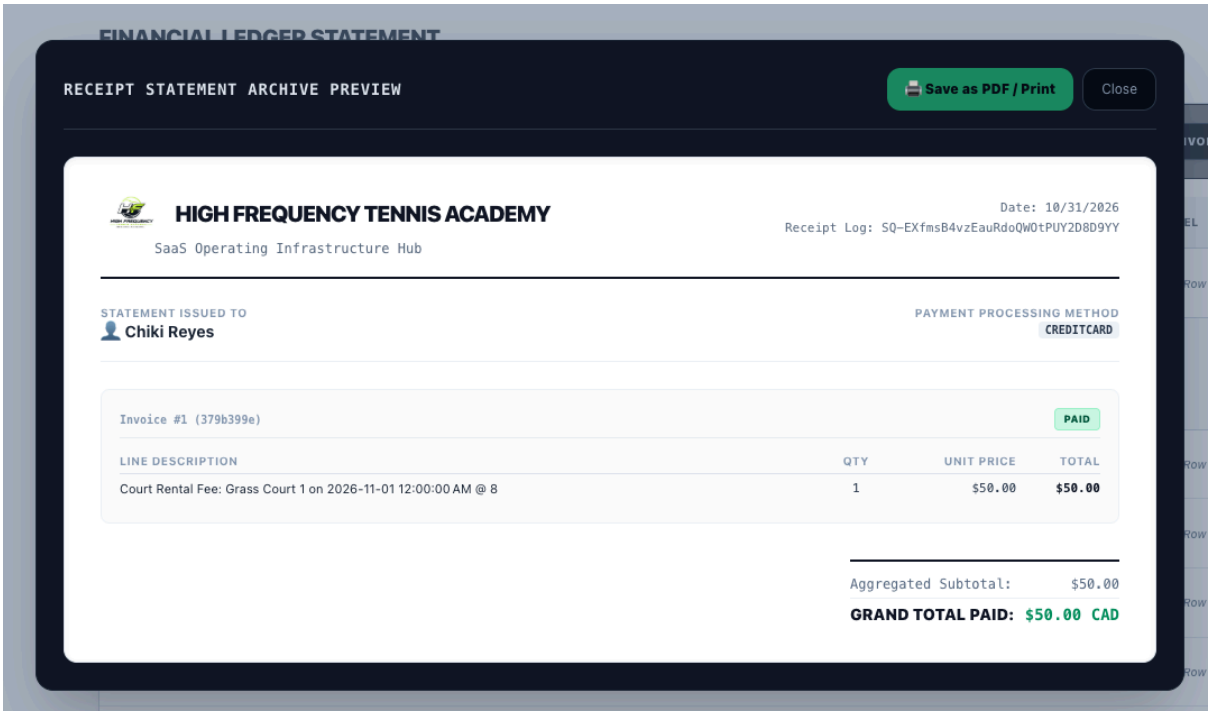
Virtualization & Infrastructure Hierarchy

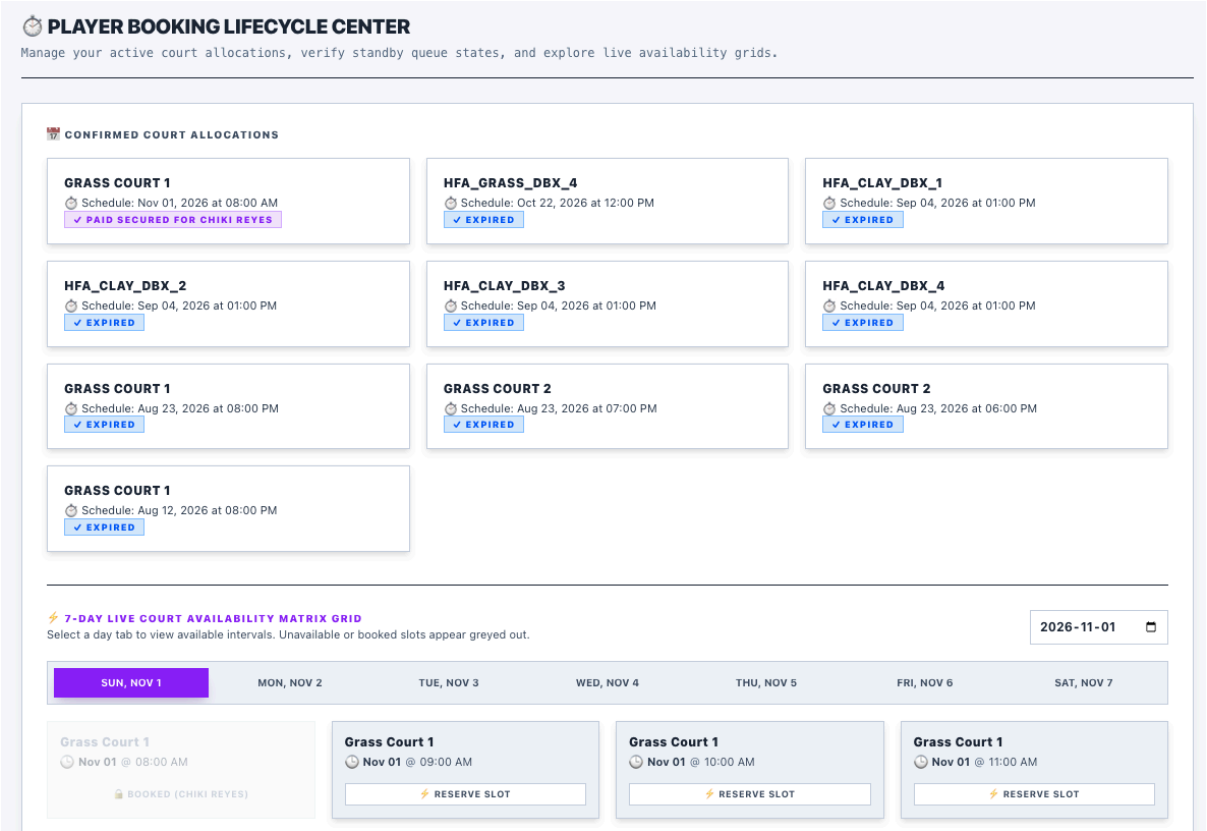




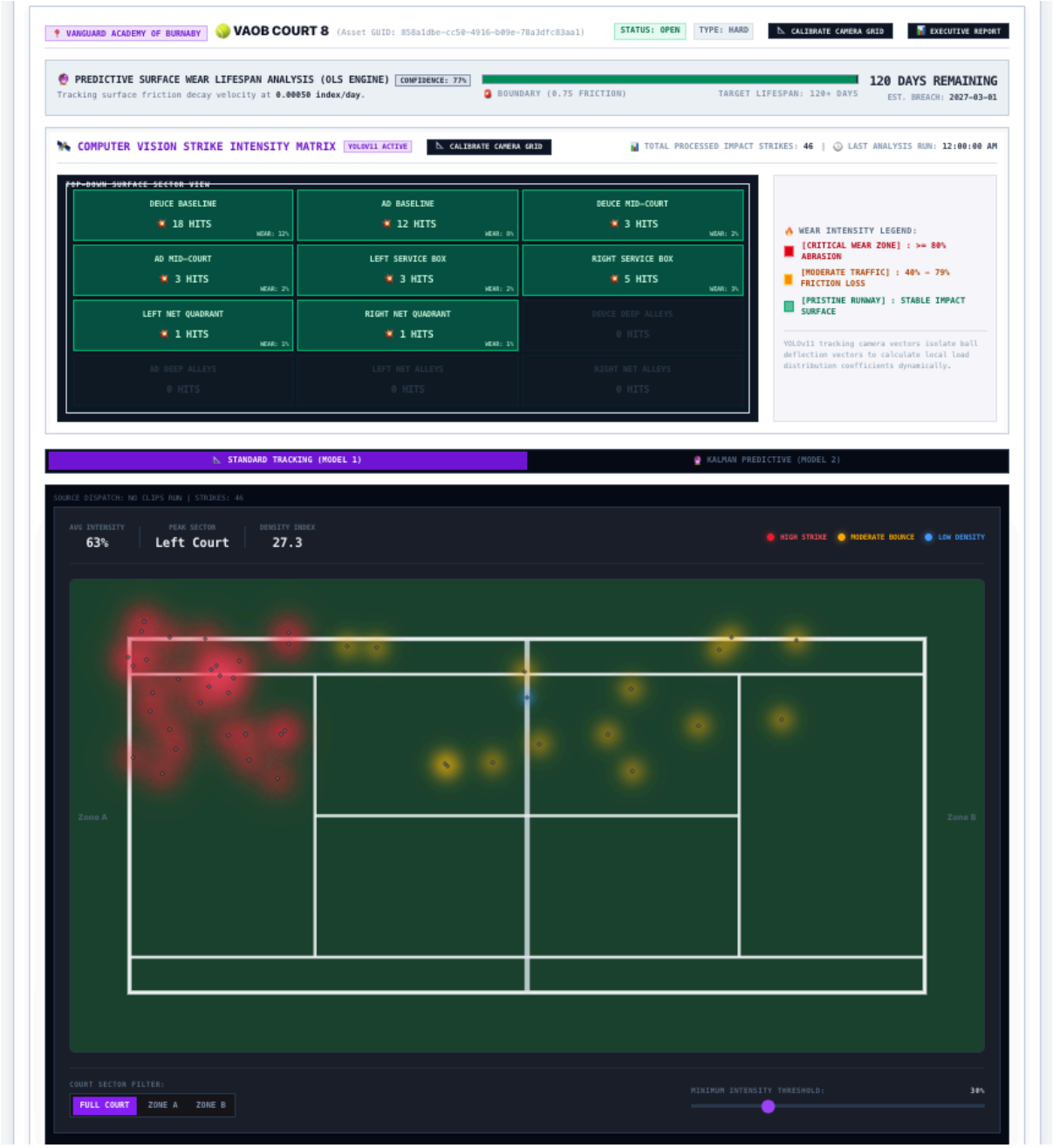
Integrated Unified Ledger & Payment Gateways







Edge-Computed Computer Vision & Trajectory Estimation



Executive Report Generation & Capstone Analytics

■ GLOBAL ACADEMY EXECUTIVE FACILITY SAFETY AUDIT

Asset identifier: VAOB Court 8 (Main Academy Facility) I Live Structural Telemetry Sheet

This executive structural evaluation audit compiles continuous multi-tenant camera metrics captured natively over port channels. During the active reporting timeframe sequence loops, the system isolated true ground impact events across the target playing coordinates fields surfaces. Our advanced predictive Kalman Filter array engine successfully reclaimed motion-blurred tracking frames, increasing data density.

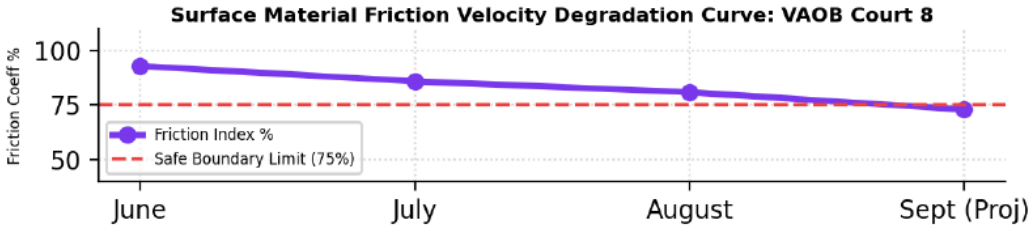
■ PREDICTIVE SURFACE WEAR LIFESPAN ANALYSIS (ADVANCED ENGINE)

Tracking Decay Velocity: 0.00053 index/day	**Target Lifespan:** 120+ Days
Operational Lifespan: 55 DAYS REMAINING	**Estimated Safety Breach: 2026-10-28**

■■ EST. BREACH NOTICE: Predictive models project an operational safety boundary breach (0.75 Friction) on 2026-10-28. Proactive court resurfacing remediation scheduling is recommended.

■ REAL-TIME FIELD SENSOR TELEMETWS DECK

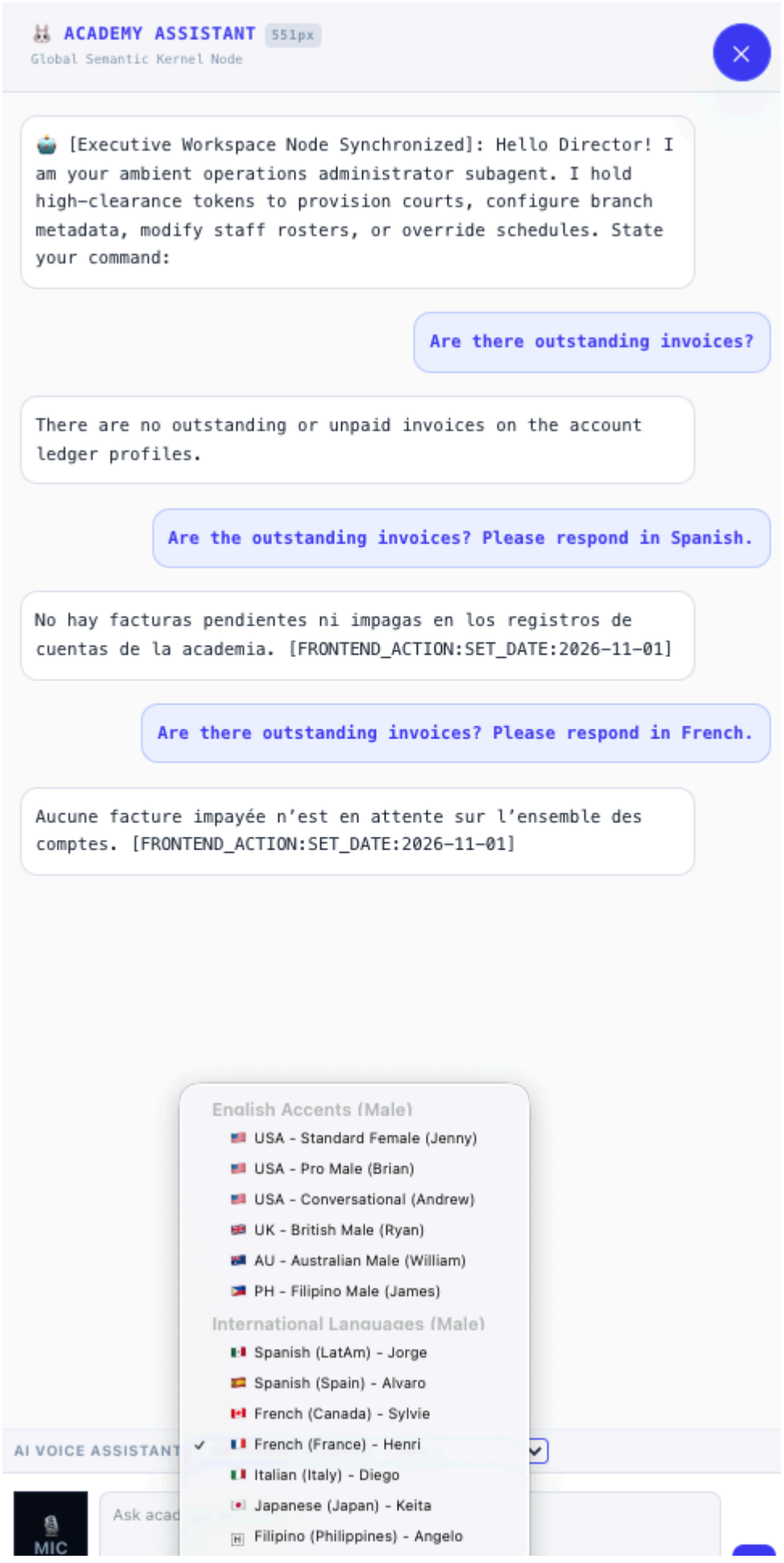
FRICTION INDEX	BOUNCE UNIFORMITY	NET TENSION	AMBIENT TEMP	HUMIDITY
0.81	97.5%	437.7 N	31.1°C	62.1%



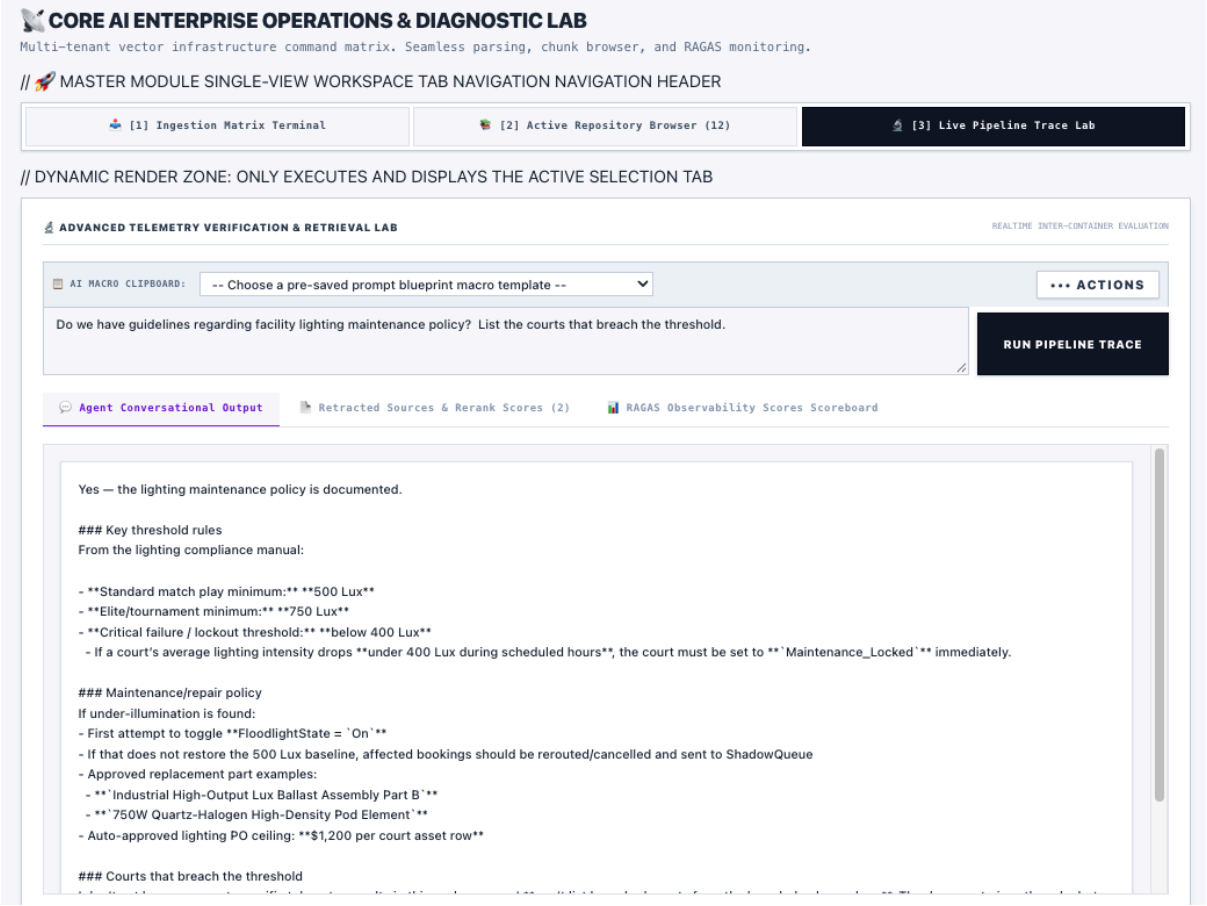
■■ LOCALIZED ANOMALY INSPECTION AND DISPATCH CLUSTERING LEDGER

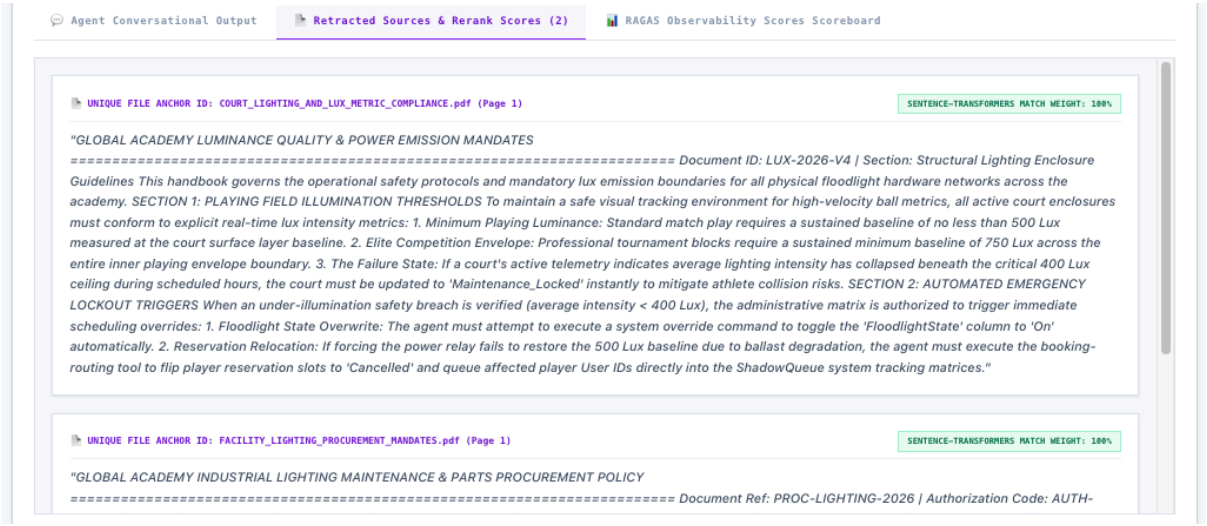
COMPONENT CATEGORY	SEVERITY LEVEL	STRUCTURAL ANOMALY RISK DESCRIPTION
Complete Facility Check	PRISTINE	All structural components evaluate optimal. No active anomalies tracked.

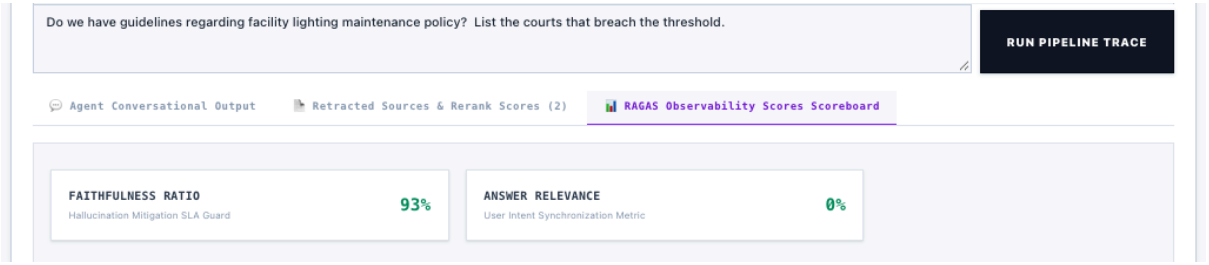
Symmetrical Voice AI Orchestration & Interface


ACADEMY ASSISTANT
551px
Global Semantic Kernel Node
[Executive Workspace Node Synchronized]: Hello Director! I am your ambient operations administrator subagent. I hold high-clearance tokens to provision courts, configure branch metadata, modify staff rosters, or override schedules. State your command:
Are there outstanding invoices?
There are no outstanding or unpaid invoices on the account ledger profiles.
Are the outstanding invoices? Please respond in Spanish.
No hay facturas pendientes ni impagas en los registros de cuentas de la academia. [FRONTEND_ACTION:SET_DATE:2026-11-01]
Are there outstanding invoices? Please respond in French.
Aucune facture impayée n'est en attente sur l'ensemble des comptes. [FRONTEND_ACTION:SET_DATE:2026-11-01]
English Accents (Male)
USA - Standard Female (Jenny)
USA - Pro Male (Brian)
USA - Conversational (Andrew)
UK - British Male (Ryan)
AU - Australian Male (William)
PH - Filipino Male (James)
International Languages (Male)
Spanish (LatAm) - Jorge
Spanish (Spain) - Alvaro
French (Canada) - Sylvie
French (France) - Henri
Italian (Italy) - Diego
Japanese (Japan) - Keita
Filipino (Philippines) - Angelo
AI VOICE ASSISTANT
MIC
Ask acad

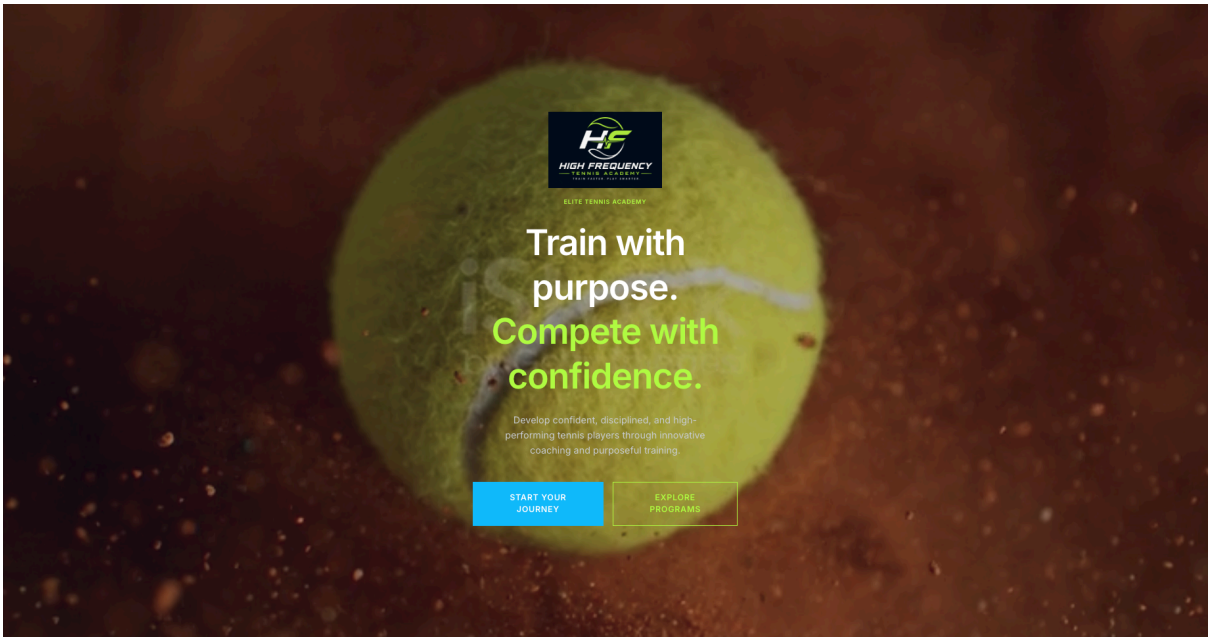
Autonomous Agentic AI Workflows & Multi-Tenant Knowledge RAG

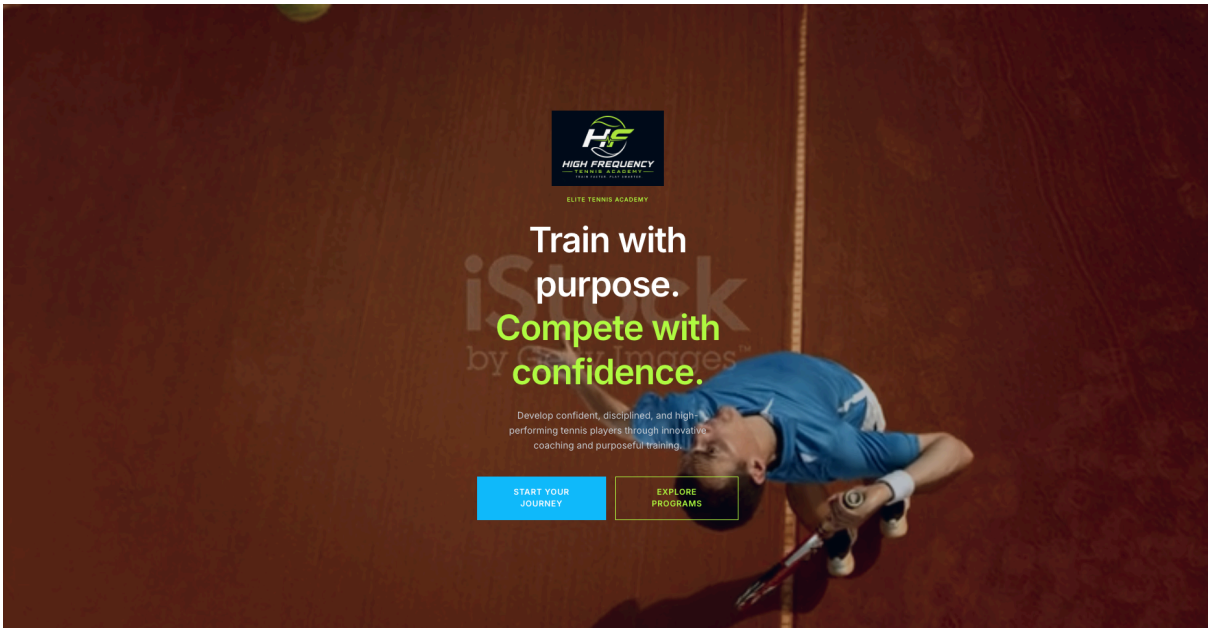


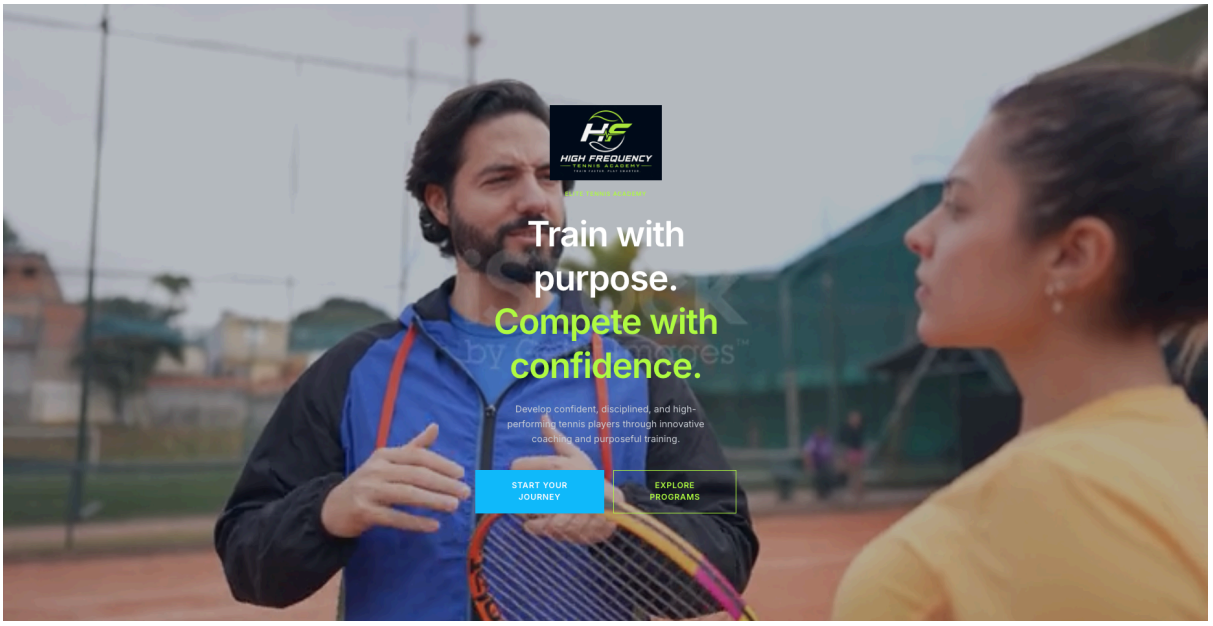




WEBSITE LANDING PAGE






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Compete with confidence.
Develop confident, disciplined, and high-performing tennis players through innovative coaching and purposeful training.
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